# OpenClaw OLED Display Tutorial

> Before starting, please complete the following in the `Pre-use Preparation` directory: [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [05-AI Voice Interaction Configuration], and [06-Three Dialogue Configuration Tests]. If you are currently only using TUI / WhatsApp, you can skip `05` for now. This article assumes the basic environment is already prepared and only covers OLED-related content.

## 1. Tutorial Objective

This article demonstrates OpenClaw's natural language control capabilities through the OLED peripheral. After completing this tutorial, readers will be able to:

- Understand the basic approach of OpenClaw controlling OLED display
- Interact with OpenClaw through three methods: WhatsApp, TUI, and voice interaction
- Complete four OLED cases: text display, metric display, weather card, and animation display
- Understand the complete chain from "user input" to "screen display"

## 2. Hardware Preparation

- OpenClaw main control board
- OLED display `[Default SSD1306 128x32 / I2C address 0x3C]`
- `[Optional] Microphone module`
- Wiring diagram




## 3. Software Preparation

**Code entry points used in this tutorial**

- Low-level driver code: `/home/jetson/hardware_hal_demo/`
- OLED control commands: `/home/jetson/hardware_hal_demo/oled`
- Temperature and humidity linked display: `/home/jetson/hardware_hal_demo/dht_oled`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`

The `oled` tool supports the following capabilities:

- Display text `text`
- Display image `image`

- Display animation `animation`
- Display weather card `weather`
- Clear screen `clear`

If you just want to quickly display temperature and humidity readings on the OLED, you can directly use `dht_oled`.

```
dht_oled  #Display temperature and humidity data on OLED
oled text --text 'Hello, World' #Display text
oled weather --location 'Shenzhen' --condition 'Sunny' --temperature '26°C' --humidity '57%'
#Weather card
oled image --path ~/hardware_hal_demo/assets/oled/mountain_photo_128x32.png #Display image
oled animation --path  ~/hardware_hal_demo/assets/oled/sunrise_anim_128x32.gif #Display GIF
animation
oled clear   # Clear screen
```

**Note: For GIF and image display, you can ask AI to generate 128x32 resolution files and place them in the path ~/hardware_hal_demo/assets/oled/**

For more commands, refer to the README documentation in this folder:

```
/home/jetson/hardware_hal_demo/README.md
```

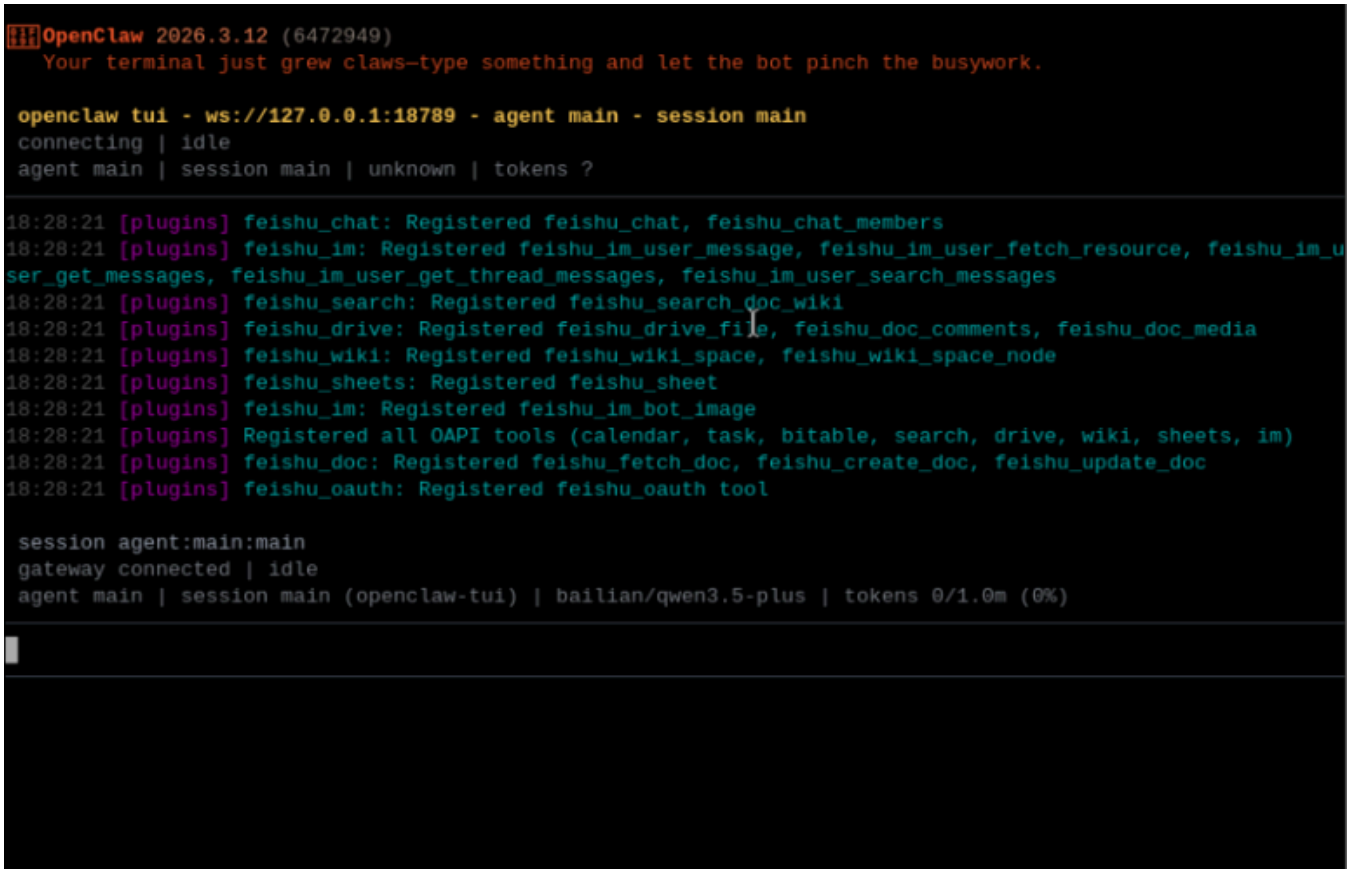
If the low-level commands execute normally, it means the hardware and the Jetson Orin NX main board are connected and communicating properly.

# 4. Testing OLED Display

Before starting this tutorial, please complete a basic `openclaw tui` dialogue self-check first; if you haven't done so, see [06-Three Dialogue Configuration Tests]. After confirming that OpenClaw can reply normally, proceed to the OLED-specific test.

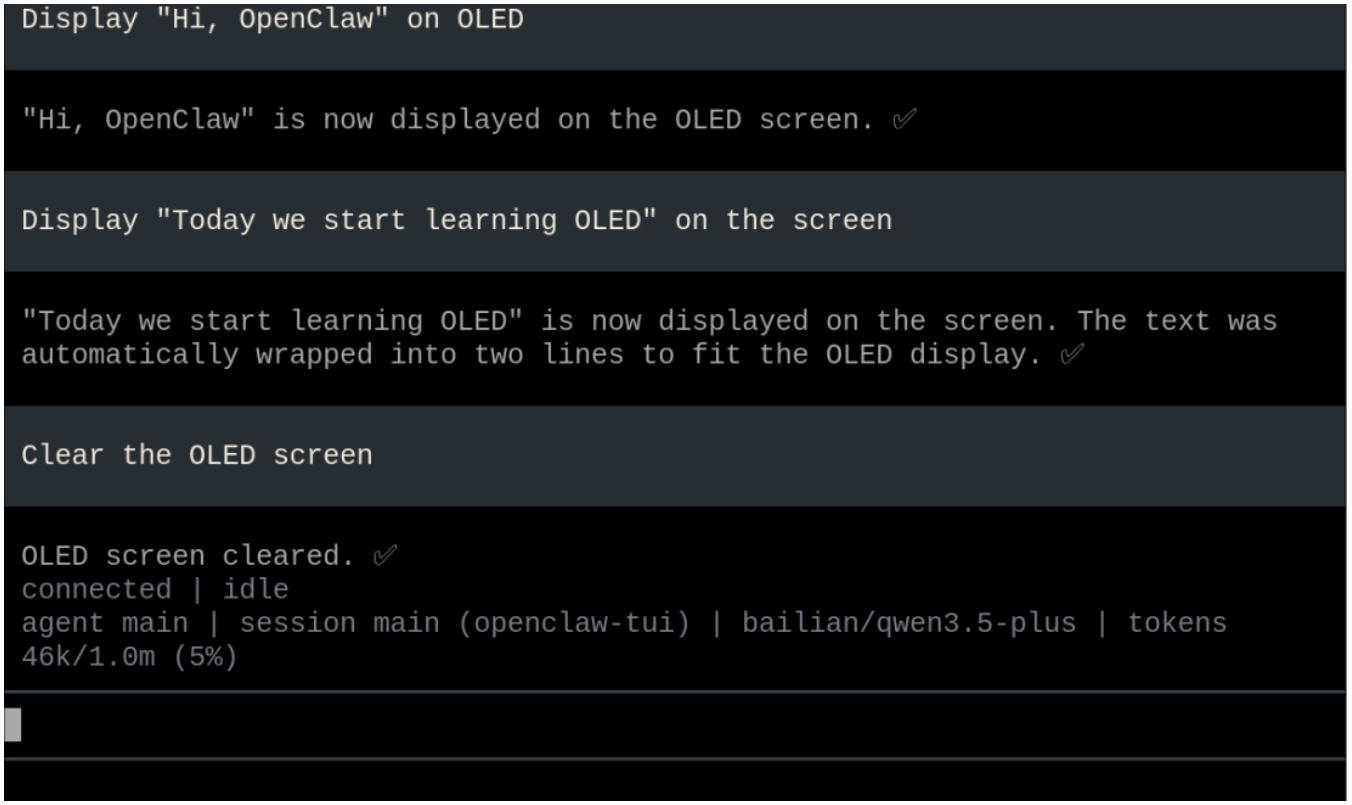
Enter the following command in the terminal to enter TUI:

```
openclaw tui
```




You can try entering the following content in the terminal:

- Display "Hello, OpenClaw" on the OLED
- Display "Starting to learn OLED today" on the screen
- Clear the OLED screen
- Display the current status "System OK" on the OLED



# 5. Three Schemes for Communicating with OpenClaw

| Scheme | Suitable Scenarios | Advantages | Recommended Use |
| --- | --- | --- | --- |
| WhatsApp | Remote control, classroom demonstration | Easy to connect, suitable for multi-person experience | Demonstrate "remote natural language control" |
| TUI | Local debugging, command verification | Most direct, easiest to troubleshoot | First run through the control chain |
| Voice code | Voice interaction, complete AI experience | Closest to real conversation experience | As the final demonstration scheme |

**If the optional AI voice module + speaker is selected, Feishu dialogue, TUI, and WhatsApp dialogue need to play back OpenClaw's replies via TTS. You can run the following program in the terminal. The AI voice module dialogue itself does not need to run it again.**

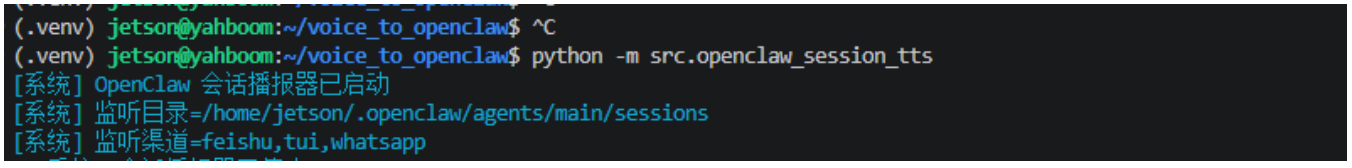
- Modify the TTS broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true  # This parameter controls whether the replies are broadcast. After modification, save it and run the following program in the terminal
```

- Enter the following in the terminal

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

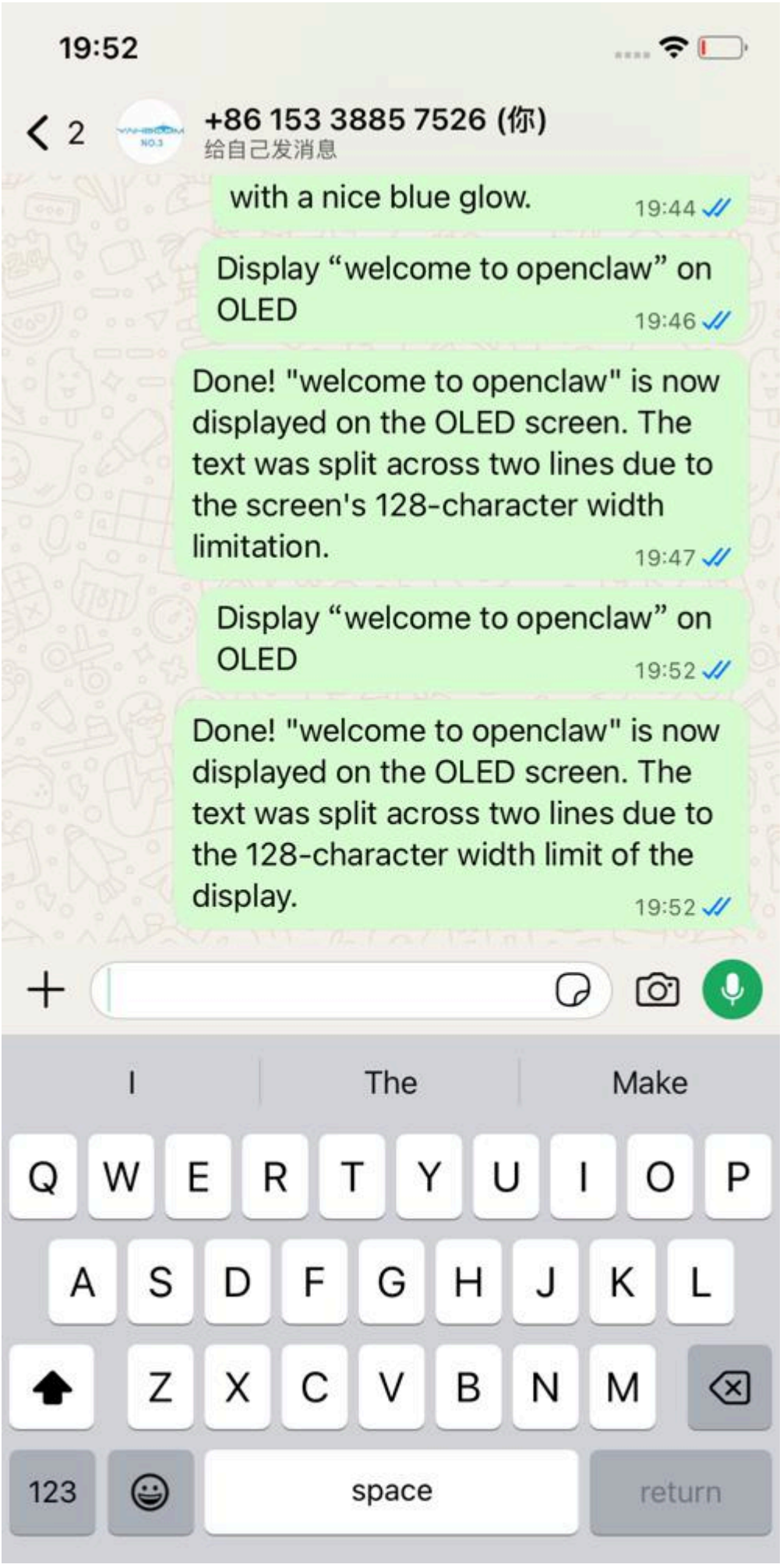



## 5.1 whatsapp Scheme

For WhatsApp connection steps, please directly refer to [06-Three Dialogue Configuration Tests]. This section only includes dialogue examples suitable for the OLED tutorial.

After creation, you can directly converse, for example:

- Display "Welcome to OpenClaw" on the OLED
- Display the current temperature and humidity data on the OLED
- Clear the OLED screen


19:52
2
+86 153 3885 7526 (你)
给自己发消息
with a nice blue glow.
19:44
Display “welcome to openclaw” on OLED
19:46
Done! "welcome to openclaw" is now displayed on the OLED screen. The text was split across two lines due to the screen's 128-character width limitation.
19:47
Display “welcome to openclaw” on OLED
19:52
Done! "welcome to openclaw" is now displayed on the OLED screen. The text was split across two lines due to the 128-character width limit of the display.
19:52
I
The
Make
Q W E R T Y U I O P
A S D F G H J K L
Z X C V B N M
123
space
return



## 5.2 TUI Scheme

If you have completed the TUI self-check in Part 0, simply enter the following command in the terminal to continue the conversation:

```
openclaw tui
```

## 5.3 Voice Code Scheme

```
Requires optional AI voice interaction module + speaker
```

For voice module installation, environment configuration, and basic wake-up process, please directly refer to [05-AI Voice Interaction Configuration]. After wake-up, you can issue voice commands directly, for example:

- Display hello on the OLED
- Change the screen content to "Welcome to OpenClaw"
- Clear the screen

```
2026-05-14 17:25:01 [INFO] voice_to_openclaw:run:26 - voice_to_openclaw starting
[系统] voice_to_openclaw 已启动
[系统] ASR=xunfei_ws/iat/en_us | 发布=gateway_chat | TTS=开/xunfei_ws | 唤醒=串口
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说"清空上下文"可以重置当前会话
[唤醒] 检测到串口信号，开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 8520 ms 语音，正在转写

[你] Display high on old.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Done! "high" is now displayed on the OLED screen.
[唤醒] 检测到串口信号，开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音，正在转写

[你] Clear the screen.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Done! The OLED screen has been cleared.
```

# 6. Comprehensive Cases

Note: The following are all demonstrated using the `openclaw tui` method. You can also choose WhatsApp or voice methods to complete them.

## 6.1 Case 1: Text Display

### Experimental Objective

Make the OLED display a fixed text to complete the most basic screen control.

### Effect Description

After OpenClaw receives the instruction, it will render the specified text on the OLED screen. This case is suitable for verifying whether text rendering, line breaks, and screen refresh are working properly.

### Example Instructions

- Display "Hello, OpenClaw" on the OLED
- Display "Today's task is to test OLED"
- Change the screen content to "System Ready"

## 6.2 Case 2: Metric Display

### Experimental Objective

Display single-line status metrics on the OLED in a compact format.

### Effect Description

This mode is suitable for compressing short information such as temperature, humidity, battery level, and status values into a single line, more like a simple dashboard reading.

### Example Instructions

- Display "Temp 26C Hum 57%" on the OLED
- Write the current status as a short single-line message
- Change the screen content to "CPU OK RAM OK"

## 6.3 Case 3: Weather Card

### Experimental Objective

Make the OLED display a more complete weather information card.

### Effect Description

The screen can display information such as location, weather, temperature, and humidity. This case is suitable for building weather stations, environmental monitors, or information panels.

### Example Instructions

- Display Shenzhen sunny 26 degrees humidity 57% on the OLED
- Create a simple weather card
- Format the current temperature and humidity into a card-style display

## 6.4 Case 4: Animation Display

### Experimental Objective

Make the OLED display images or animations to display richer visual effects.

### Effect Description

In addition to static text, the OLED can also display bitmaps, GIF animations, or simple self-check animations. This case is great for boot screens, emoji animations, or classroom demonstrations (you can prepare your own photos or GIF animations to display and place them in the path /home/jetson/hardware_hal_demo/assets/oled).

### Example Instructions

- Play a simple animation on the OLED
- Display an icon image
- Create a boot welcome animation